

Assignment 1

Problem Statement

In software development, stakeholders often communicate requirements through meetings, discussions, and interviews. Manually converting these conversations into structured requirements is time-consuming and may lead to missing or ambiguous requirements.

Develop an AI-powered system that accepts an uploaded audio/video recording, converts it into text, and automatically generates:

- Functional Requirements (FR)
- Non-Functional Requirements (NFR)

The system should analyze the conversation intelligently and produce clear requirement statements.

Objective

The objective of this assignment is to:

- Understand AI-assisted Requirement Engineering
 - Convert speech recordings into transcripts
 - Use Generative AI/LLMs for requirement extraction
 - Automatically identify and classify Functional and Non-Functional Requirements
 - Generate structured software requirement outputs
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Features to Implement

1. Upload audio/video recording (.mp3, .wav, .mp4)
2. Convert speech to text using Speech-to-Text models/APIs
3. Clean and process generated transcripts
4. Extract Functional Requirements (FR)
5. Extract Non-Functional Requirements (NFR)
6. Categorize NFRs (Security, Performance, Reliability, etc.)
7. Display generated requirements in structured format
8. Export generated requirements as PDF/DOCX/TXT